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6880/01

For examination from 2027 - 2029

1 hour

Additional materials required: Scientific calculator
Geometrical Instruments

The total number of marks for this paper is 60.

[Turn over

- 1 List all prime factors of 30.

Answer [2]

- 2 Tandzile has a cat.

She feeds her cat $\frac{2}{3}$ litre of milk each day.

Milk is sold in 500 ml cartons in a shop.

How many cartons must she buy to feed her cat for 10 days?

Answercartons [3]

- 3 Without using a calculator, work out.

$$0.63 \div 0.7$$

You must show all your working.

Answer [2]

- 4 Express $\frac{27}{40}$ as a percentage.

Answer % [2]

5 Express each of the following numbers as a power of 5.

(a) 125

Answer (a)..... [1]

(b) 0.2

Answer (b) [1]

(c) 1

Answer (c) [1]

6 There are 30 learners in a class.

24 learners study Biology.

19 learners study Chemistry.

16 learners study both Biology and Chemistry.

(a) Draw a Venn diagram showing this information.

[3]

(b) How many learners do not study any of the two subjects?

Answer (b).....learners [1]

7 Factorise completely.

(a) $81p - 18$,

Answer (a) [1]

(b) $36 - 4x^2$.

Answer (b) [2]

8 Triangle EFG is such that $\hat{EFG} = 53^\circ$, $EF = 7$ cm and $FG = 5$ cm.

(a) Draw accurately triangle EFG .

[3]

(b) Measure and write down the length of EG .

Answer (b)cm [1]

- 9 The following are scores of 11 learners in a mathematics examination.

45 66 38 56 85 63 24 13 69 41

- (a) Calculate the mean.

Answer (a) [3]

- (b) Calculate the range.

Answer (b) [1]

-
- 10 Tanele is x years old.
Her brother is 2 years older than her.
Their mother is five times as old as Tanele.
The sum of all their ages is 79 years.

- (a) Express the brother's age in terms of x .

Answer (a) [1]

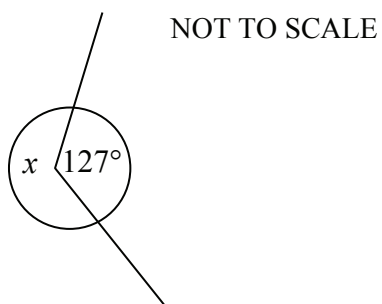
- (b) Form an equation in terms of x .

Answer (b) [1]

- (c) Solve the equation to find Tanele's age.

Answer (c) years [2]

11



Calculate the size of the angle x .

Answer $x = \dots\dots\dots^\circ$ [1]

12 Given that $a = 0.23$ and $b = 1.7$, find

(a) ab^2 ,

Answer (a)..... [1]

(b) $(3a^2) - 2b^2$,

Answer (b)..... [1]

(c) $\frac{a+b}{a-b}$.

Answer (c) [1]

- 13 (a)** Calculate the area of a circle of radius 2.2 m.

Answer (a) m² [2]

- (b)** A triangle has sides of lengths 3.5 cm, 12 cm and 12.5 cm.

Is the triangle right-angled? Justify your answer.

Answer (b).....

.....

..... [2]

- 14** The length of a side of a square is 9 cm.

- (a)** Find the area of the square.

Answer (a)..... cm² [1]

- (b)** Calculate the length of the diagonal.

Answer (b)..... cm [2]

17 You are given that $\overrightarrow{PR} = \begin{pmatrix} 8 \\ -4 \end{pmatrix}$ and $\overrightarrow{RS} = \begin{pmatrix} 1 \\ 9 \end{pmatrix}$

(a) Find $-2\overrightarrow{PR}$.

Answer (a)..... [1]

(b) Express $\overrightarrow{PS}$ as a column vector

Answer (b)..... [1]

18 (a) Simplify $7 - 3(4 - 5d)$.

Answer (a)..... [2]

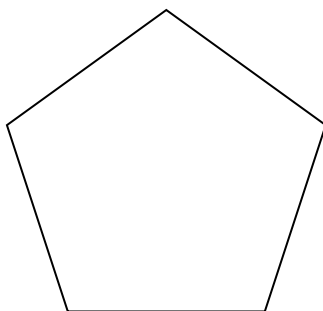
(b) Express as a single fraction in its simplest form $\frac{4-r}{5} + \frac{7r-1}{15}$.

Answer (b)..... [3]

- 19 You are given that $M = \begin{pmatrix} -1 & 3 \end{pmatrix}$ and $N = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$.
Calculate NM .

Answer [2]

- 20 The diagram shows a regular pentagon.



NOT TO SCALE

- (a) Write down the sum of its exterior angles.

Answer (a) [1]

- (b) Calculate the size of an interior angle of the pentagon.

Answer (b) [2]

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